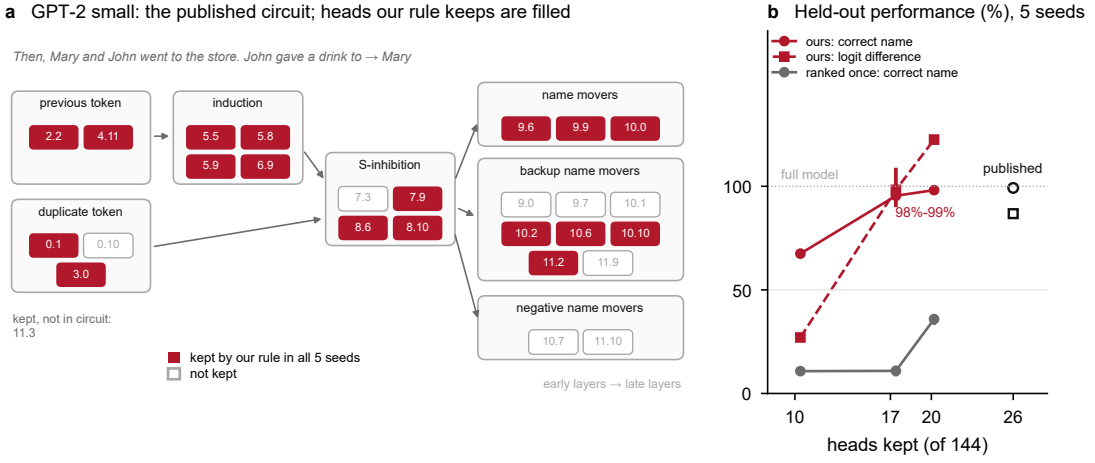

a GPT-2 small: the published circuit; heads our rule keeps are filled
Then, Mary and John went to the store. John gave a drink to → Mary
previous token
2.2
4.11
induction
5.5
5.8
5.9
6.9
duplicate token
0.1
0.10
3.0
S-inhibition
7.3
7.9
8.6
8.10
name movers
9.6
9.9
10.0
backup name movers
9.0
9.7
10.1
10.2
10.6
10.10
11.2
11.9
negative name movers
10.7
11.10
kept, not in circuit:
11.3
kept by our rule in all 5 seeds
not kept
early layers → late layers
b Held-out performance (%), 5 seeds
ours: correct name
ours: logit difference
ranked once: correct name
full model
published
98%-99%
100
50
0
10
17
20
26
heads kept (of 144)